

Sari Sadiya, PhD

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Principal Investigator at Constructor Labs. Dual PhD in Computer Science and Cognitive Neuroscience with a mathematics background. Once a VLSI engineer at Apple. Focusing on machine learning for bioinformatics, neuroscience, and computer vision.

PROFESSIONAL EXPERIENCE

Constructor Knowledge Labs

Bremen, Germany

Principle Investigator

June 2026–Present

- Human-AI Fusion & Cognitive Wearable Intelligence project

Frankfurt Goethe University

Frankfurt, Germany

Research Associate at Prof. Roig's Computational Vision and Artificial Intelligence lab

Dec 2024–Present

- Awarded a Hessian.AI Grant "Cognitive Neural Architecture Search" to lead a research group focusing on identifying artificial neural network architectures with brain-like organizational properties
- Course instructor and thesis supervisor

Machine Learning Consultant

Remote

Built ML systems and infrastructure for startups of various growth stages, focusing on full-stack and backend solutions using modern frameworks and cloud tools

Jan 2023–Dec 2024

- **Jience** (2024): Developed a recommendation system for a job matching platform
Tech stack: Next.js, Flask, Python, MongoDB, Weaviate, Google Cloud Functions
- **Aneesi.ai** (2023): Developed an agentic b2b marketing system agent.
Tech stack: Flask, Python, Firestore, Weaviate, SQL, Google Cloud Functions
- **GamerTag** (2023): Implemented an embedding-based social recommender system
Tech stack: Python, Zilliz, AWS

Frankfurt Institute for Advanced Studies

Frankfurt, Germany

Research Fellow

Nov 2022–Nov 2024

- Co-advised by Prof. Cichy at the FU Berlin and Prof. Roig at Frankfurt Goethe University
- Course Instructor: Taught 4 semesters of "Pattern Analysis and Machine Intelligence Seminar"
- Thesis Supervisor: Mentored 9 junior researchers, resulting in several conference publications

Apple

Haifa, Israel

Engineer

Mar 2013–Aug

- Embedded development, Circuit design and verification with Verilog and systemVerilog Assertions.

EDUCATION

MICHIGAN STATE UNIVERSITY

Dual PhD in Computer Science and Cognitive Neuroscience

Thesis: Machine Learning Approaches for Processing and Decoding Attention Modulation of Sensory Representations from EEG (GPA: 3.85/4.0)

Michigan, USA

Oct 2022

MICHIGAN STATE UNIVERSITY (Fulbright IIE Scholarship)

Masters in Engineering with focus on Natural Language Processing

Thesis: Interactive Human-Robot One-Shot Task Structure Learning from Demonstration (GPA: 3.75/4.0)

Michigan, USA

Aug 2017

TECHNION - ISRAEL INSTITUTE OF TECHNOLOGY

Bachelors of Science in Mathematics (GPA equivalent to: 3.65/4.0)

Haifa, Israel

Jun 2013

TECHNION - ISRAEL INSTITUTE OF TECHNOLOGY

Bachelors of Science in Computer Science (GPA equivalent to: 3.65/4.0)

Haifa, Israel

Jun 2013

ADDITIONAL SKILLS

- **Machine Learning:** Extensive experience training and fine-tuning convolutional and transformer models
PyTorch, Sklearn, Numpy, Pandas, OpenCV, Slurm, MLflow, C++
- **Full Stack Development:** I build accessible applications to carry my work beyond the walls of academia
RestfulAPIs, VueJS, Bootstrap, PyQt, Flask, MongoDB, SQL
- **Experimental Psychology:** Designing and conducting behavioral, eye-tracking, EEG, and fMRI experiments
Matlab, FreeSurfer, EEGLAB, PsychoPy, E-Prime, SPSS
- **Hardware and Embedded Development:** When the hardware I need does not exist I build it
Verilog and SystemVerilog Assertions, x86 Assembly

LANGUAGES

- English (Full Working Proficiency)
- German (CEFR A2)
- Arabic (Native Speaker)
- Hebrew (Native Speaker)

DISTINCTIONS AND AWARDS

- **Dissertation Completion Fellowship** MSU Graduate Office 2021
- **Cultural Heritage Fellowship** Matrix center for digital humanities & social sciences 2020
- **Graduate Office Fellowship** MSU Graduate Office 2020
- **Fulbright Scholarship Program for International Educational Exchange** 2015-2017
Granted by the U.S. Bureau of Educational and Cultural Affairs.
- **Technion Honor Role** Awarded to top 10% of students in each program 2013

ACADEMIC GRANTS

<i>Human-AI Fusion & Cognitive Wearable Intelligence</i> by Constructor Tech	850,000 €	2026-29
<i>Cognitive Neural Systems Architecture Search</i> by Hessian.AI	100,000 €	2025

ADDITION EXPERIENCE

Project director for the Refugee Outreach Collective “Global Classroom”, a partnership with Central Michigan University. The project expands educational access to individuals experiencing displacement at Malawi Dzaleka refugee camp through participation in accredited online classes.

PUBLICATIONS

My research is at the intersection of Computer Science and Cognitive Neuroscience. Generally, conference proceedings are considered more prestigious in computer science, while journal publications are preferred in psychology. * marks joint first authorship with students I supervised, or the corresponding author position.

Computer Science Publications

- A. Hohmann, G. Roig*, **S. Sadiya***. *Language Model Architecture Shapes Brain Alignment and Neuroscientific Inference*. In Springer Lecture Notes on Artificial Intelligence (Originally Oral ANNPR 2026)
- M. Ahsan, **S. Sadiya***, G. Roig*. *Investigating Spurious Cue Sensitivity in Semantic Segmentation*. In Springer Lecture Notes on Artificial Intelligence (Originally Oral ANNPR 2026)
- M. Guo, B. Choksi, **S. Sadiya**, P. Oyarzo, R. Cichy, G. Roig; *Predictive Coding inspired convolutional networks can capture the neural dynamics of recurrent processing in human image recognition*. In Proceedings of the European Symposium on Artificial Neural Networks (Oral ESANN 2026)
- M. Ahsan, D. Tawadros, **S. Sadiya**, L. Quynh, J. Schlotterer, C. Seifert, G. Roig. *Weakly Supervised Shortcut Mitigation Using Sparse Projections*. In Proceedings of the European Symposium on Artificial Neural Networks (Oral ESANN 2026)
- L. Kuhn*, **S. Sadiya***, J. Schlotterer, F. Buttner, C. Seifert, G. Roig. *Efficient unsupervised shortcut learning detection and mitigation in transformers*. In Proceedings of the 2025 International Conference on Computer Vision (ICCV 2025)
- L. Kuhn*, **S. Sadiya***, G. Roig; *Cognitive Neural Architecture Search Reveals Hierarchical Entailment*. In the ICLR Second Workshop on Representational Alignment (2025)
- D. Bersch, M. Vilas, **S. Sadiya**, T. Schaumloffel, K. Dwivedi, C. Sartzetaki, R. Cichy, G. Roig. *Net2Brain: A Toolbox to Compare Artificial Vision Models with Human Brain Responses*. In Frontiers in Neuroinformatics Volume 19 (2025)
- M. Guo, M. Samjatin, B. Choksi, **S. Sadiya**, R. Cichy, G. Roig; *Predictive Coding Dynamics Enhance Model-Brain Similarity*. In Proceedings of the European Symposium on Artificial Neural Networks (ESANN 2025)
- M. Sayed, **S. Sadiya**, J. Wichtlhuber, J. Dietz, M. Neitzel, L. Keller, G. Roig, A. Bucher; *Evaluating Medical Image Segmentation Models using Augmentation*. In Tomography (2024)
- M. Guo, B. Choksi, **S. Sadiya**, A. Gifford, M. Vilas, R. Cichy, G. Roig; *Limited but consistent gains in adversarial robustness by co-training object recognition models with human EEG*. (Oral In the ECCV Workshop on Human-inspired Computer Vision 2024)
- R. Khanmohammadi, **S. Sadiya**, S. Esfandiarpour, T. Alhanai, M. Ghassemi; *MambaNet: A Hybrid Neural Network for Predicting the NBA Playoffs*. In Springer Nature Computer Science (2024)

- T. Ettling*, **S. Sadiya***, G. Roig; *Different Algorithms (Might) Uncover Different Patterns: A Brain-Age Prediction Case Study*. In Proceedings of the IEEE International Conference on Bioinformatics and Biomedicine (Oral BIBM 2023)
- **S. Sadiya**, T. Alhanai, M. Ghassemi; *Feature Imitating Networks*. In Proceedings of the IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP 2022).
- A. Akula, K. Wang, C. Liu, **S. Sadiya**, H. Lu, S. Todorovic, J. Chai, S. Zhu; *CX-ToM: Counterfactual explanations with theory-of-mind for enhancing human trust in image recognition models*. In Iscience (2023)
- **S. Sadiya**, T. Alhanai, M. Ghassemi; *Artifact Detection and Correction in EEG data: A Review*. In Proceedings of the 2021 IEEE International Conference on Neural Engineering (NER 2021).
- **S. Sadiya**, T. Alhanai, T. Liu, M. Ghassemi; *Unsupervised EEG Artifact Detection and Correction*. In Frontiers in Digital Health. Volume 2.
- **S. Sadiya**, T. Alhanai, T. Liu, M. Ghassemi; *EEG Channel Interpolation Using Deep Encoder-decoder Networks*. In Proceedings of the IEEE International Conference on Bioinformatics and Biomedicine (BIBM 2020).
- S. Yang, **S. Sadiya**, Q. Gao, Y. Chai; *Commonsense Action Explanation in Human-Agent Communication*. In Proceedings of the Conference on Empirical Methods in Natural Language Processing (EMNLP 2018).
- J. Chai, Q. Gao, L. She, S. Yang, **S. Sadiya**, G. Xu; *Language to Action: Towards Interactive Task Learning with Physical Agents*. In Proceedings of the International Joint Conferences on Artificial Intelligence (IJCAI 2018).
- H. Karimi, P.C. Roy, **S. Sadiya**, J. Tang; *Multi-Source Multi-Class Fake News Detection*. The 27th International Conference on Computational Linguistics (COLING 2018).
- C. Liu, S. Yang, **S. Sadiya**, N. Shuka, Y. He, S. Zhu, and J. Y. Chai; *Joint Learning Task Structures from Language Instruction and Visual Demonstration*. In Proceedings of the Conference on Empirical Methods in Natural Language Processing (EMNLP 2016).

Computational Cognitive Neuroscience Publications

- M. Menciloglu*, **S. Sadiya***, E. Chantland, M. Ghassemi, S. Revizza, T. Liu; *Limits of attention to multiple colors revealed by EEG decoding*. In Imaging Neuroscience (2025)
- Dwivedi, **S. Sadiya***, M. Balode, G. Roig, R. Cichy; *Visual features are processed before navigational affordances in the human brain*. Nature Scientific Reports (2024)
- A. Burgoyne, **S. Sadiya**, L. Harris, M. Becker, J. Brascamp, D. Hambrick; *Revisiting the Self-Generation Effect in Proofreading*. In Psychological Research: Journal of Perception, Attention, Memory, and Action (2023)
- T. Liu, M. Fang, **S. Sadiya**; *Adaptive visual selection in feature space*. Psychonomic Bulletin & Review (2022)

Preprints and Manuscripts Under Review

- **S. Sadiya***, L. Kuhn*, C. Chandra, V. Ramesh, G. Roig. *Cognitive Neural Architecture Search Identifies Robust Models with Brain Like Hierarchy* (under review at IJCV 2026)
- A. Gifford, B. Lahner, **S. Sadiya**, M. Vilas, A. Lascelles, A. Oliva, K. Kay, G. Roig, R. Cichy. *The Algonauts Project 2023 Challenge: How the Human Brain Makes Sense of Natural Scenes* (2023)

INVITED TALKS

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- Al-Qattan Cultural Centre 2025: **S. Sadiya**:
Creative Writing in the Age of Artificial Intelligence
 - Grade Brain Center at Frankfurt Goethe University 2025: **S. Sadiya**
Modelling Cognitive Functions with Artificial Neural Networks - A practical workshop
 - Cutting Gardens at Frankfurt Germany 2023: **S. Sadiya**, D. Bersch, G. Roig;
Net2Brain: Unraveling cognitive functionality as captured by EEG using pretrained artificial neural networks
 - Conference on Cognitive Computational Neuroscience at Oxford UK 2023: **S. Sadiya**, A. Gifford, M. Vilas
The 2023 Algonauts Challenge Hackathon